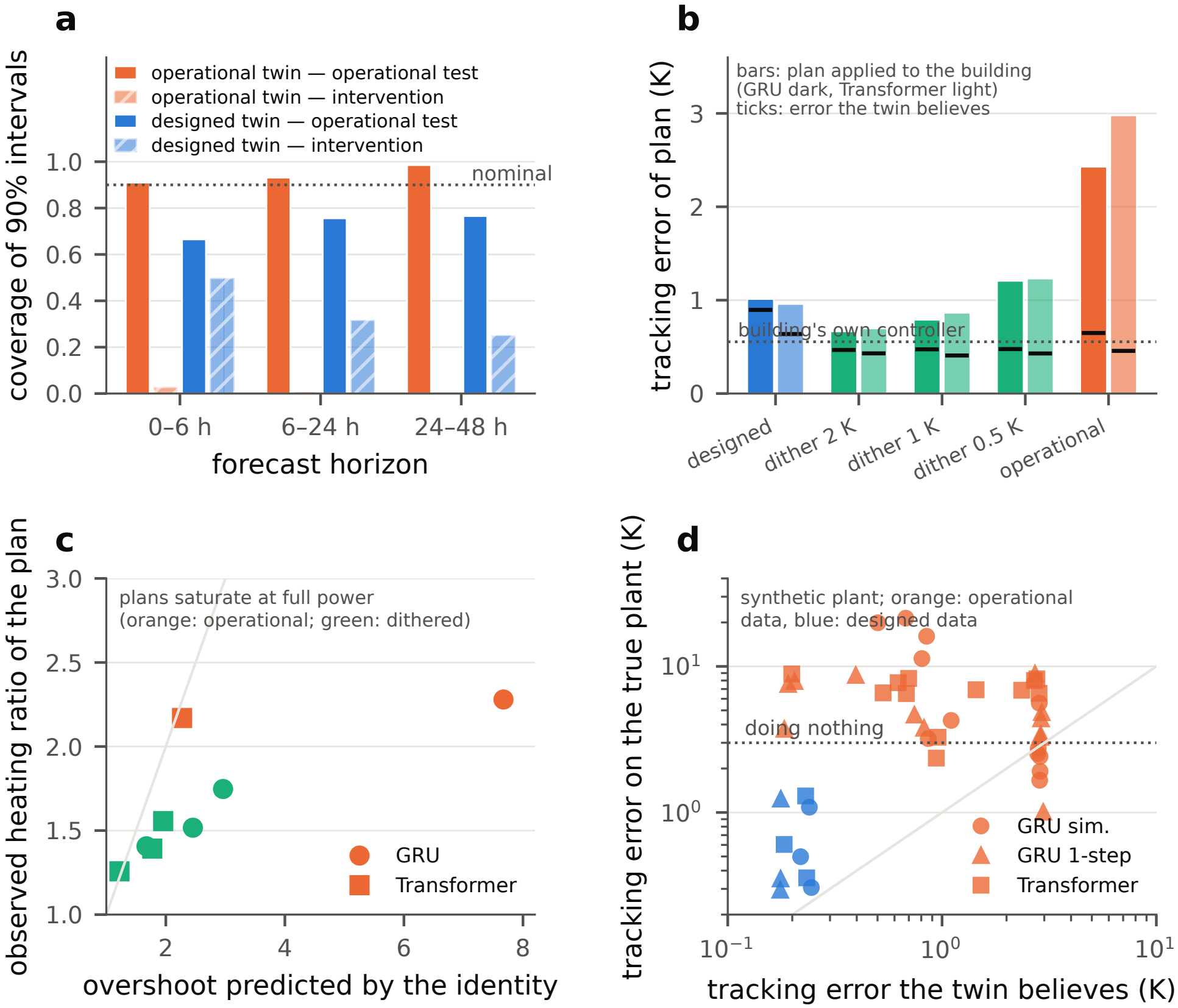
a
operational twin — operational test
operational twin — intervention
designed twin — operational test
designed twin — intervention
nominal
coverage of 90% intervals
1.0
0.8
0.6
0.4
0.2
0.0
0–6 h
6–24 h
24–48 h
forecast horizon
b
bars: plan applied to the building
(GRU dark, Transformer light)
ticks: error the twin believes
tracking error of plan (K)
3
2
1
0
building's own controller
designed
dither 2 K
dither 1 K
dither 0.5 K
operational
c
plans saturate at full power
(orange: operational; green: dithered)
observed heating ratio of the plan
3.0
2.5
2.0
1.5
1.0
2
4
6
8
GRU
Transformer
overshoot predicted by the identity
d
synthetic plant; orange: operational
data, blue: designed data
tracking error on the true plant (K)
10^1
10^0
doing nothing
GRU sim.
GRU 1-step
Transformer
10^{-1}
10^0
10^1
tracking error the twin believes (K)